

FASTQ Quality Control

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Introduction

Every high-throughput sequencing analysis starts with FASTQ quality control. Before alignment, before quantification, before any downstream interpretation, you should inspect per-base quality, adapter contamination, GC content, and duplicate rate to decide on trimming and filtering strategies. The cost of skipping this step is high: low-quality data quietly degrade alignment accuracy, inflate variant-calling false positives, and bias differential-expression conclusions in ways that are hard to detect retrospectively.

Prerequisites

A working understanding of the FASTQ format, Phred quality scores, and basic Illumina sequencing biology.

Theory

The Phred quality score is $Q = -10 \log_{10} P(\text{error})$, so Q30 means 1 in 1000 probability of base-call error and Q40 means 1 in 10,000. Modern Illumina runs aim for Q30 or higher across most of each read, with quality typically dropping toward the 3' end.

Standard FastQC-style metrics:

- **Per-base sequence quality:** boxplot of Phred scores at each position; declines toward 3' are normal but should not drop below Q20.
- **Per-sequence quality:** histogram of mean read quality; bi-modal histograms suggest mixed-quality libraries.
- **GC content:** should match expected genomic GC; deviations indicate contamination or biased library prep.
- **Adapter contamination:** presence of known adapter sequences at 3' ends.
- **Overrepresented sequences:** top recurring sequences often pinpoint adapters or contamination.

Assumptions

Modern Illumina FASTQ files with Sanger-encoded Phred (Phred+33). Older Illumina encodings (Phred+64) require explicit specification.

R Implementation

```
library(ShortRead); library(Rqc)

# Assume demo FASTQ file path
fq_file <- system.file("extdata", "ex1.fq", package = "ShortRead")

# Basic inspection
reads <- readFastq(fq_file)
length(reads)
head(width(reads))
mean(as.numeric(encoding(quality(reads)))) # mean Phred base
```

```
# Rqc for systematic QC
qc <- rqcQA(fq_file, workers = 1)
rqcReport(qc)
```

Output & Results

`Rqc::rqcQA()` produces an interactive HTML report covering per-position quality, GC content, duplicate rate, length distribution, and adapter flags. `ShortRead::readFastq()` provides programmatic access for custom diagnostics.

Interpretation

A reporting sentence: “FastQC-style QC on the 12 RNA-seq samples showed mean per-base Phred ranging from 35 to 38 across the read length, adapter contamination below 1 %, and GC content centred at 49 % (within the expected range for human transcriptome). After fastp trimming at Q15 with adapter removal, approximately 95 % of bases were retained per sample.” Always report pre- and post-trim QC.

Practical Tips

- Use `fastp` or `trim_galore` at the command line for production QC and trimming; R tools (`Rqc`, `ShortRead`) are useful for exploration but are slower than the dedicated command-line utilities on large datasets.
- Report pre- and post-trim QC; the QC improvement validates the trimming step.
- GC content far from the expected genomic value suggests contamination, library prep biases, or species mismatch; investigate before alignment.
- Overrepresented sequences often pinpoint adapter sequences, rRNA contamination, or PCR duplicates of single dominant transcripts.
- For multiple samples, `MultiQC` aggregates per-sample reports into a single dashboard; this is essential for reviewing dozens or hundreds of samples consistently.
- Phred Q20 is the conventional threshold for “good” bases; trimming at Q15 retains more data while removing the worst calls.

R Packages Used

`ShortRead` for general FASTQ manipulation in R; `Rqc` for systematic QC reports; `Rfastp` as a wrapper around `fastp` for fast trimming and QC; `ngsReports` for parsing and aggregating `MultiQC` and `FastQC` outputs into R objects.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- RNA-seq with `DESeq2` / `edgeR`
- Enrichment analysis (`GSEA` / `ORA`)
- scRNA-seq with `Seurat`